Reproducibility Manifest and Supplementary Material Checklist

This document maps the artifacts already generated across this project's development to what should be packaged and where, before submission. Convert this into the actual public repository structure described in the Data Availability Statement.

1. Source data (do not re-upload; cite and checksum-reference only)

Item	Role	Verification
`Twall-100.hdf5`, `Twall-103.hdf5`, `Twall-106.hdf5`	Tutorial-split train/val/test	SHA-256 checksums recorded in Methods §3.1
Legacy 10-trajectory archive (`Twall-79...110`)	Cross-condition split source	Archive SHA-256 recorded in Methods §3.1

Do not re-host the raw BubbleML data files themselves unless the dataset's license explicitly permits redistribution — link to the official source and cite the checksums as verification that your copy matches the canonical release.

2. Code (public GitHub repository, referenced by DOI)

Component	Contents
Data pipeline	Preprocessing, schema validation, trajectory-level split logic, HDF5 loaders for both tutorial and legacy-archive formats
Model definitions	FNO, T-FNO, F-FNO, U-Net, local-global hybrid, divergence-penalty hybrid, bounded-alpha output head
Training scripts	Full training loop, validation-plateau stopping rule, checkpoint saving/safe-loading (`weights_only=True`)
Evaluation scripts	Five-field benchmark metrics, interface/conservation metrics, autoregressive CHF-proxy rollout evaluator
Statistical analysis	Paired bootstrap, exact sign-flip test, Holm-Bonferroni correction (single unchanged implementation used throughout)
Plotting/figures	Loss curve, dry-area trace, Pareto-front, lambda-sensitivity, benchmark-workflow, and split-diagram generation scripts; field snapshots are declared unavailable
Tests	Full test suite covering architecture, serialization, and statistical protocol correctness

3. Configuration and provenance (include in repository, reference in Supplementary)

For every experiment referenced in the manuscript, the repository should retain:

- `config.yaml` (exact hyperparameters used)
- `results.json` (full training history: per-epoch train/val loss, wall time, stop reason)
- Checkpoint file with recorded SHA-256, seed, and git commit hash
- Runtime environment record (Python/PyTorch version, CUDA/MPS availability, hardware)

This is available for the locally retained tutorial and intervention experiments. The compact cross-condition export lacks complete per-seed configs, histories, and checkpoints; this gap is documented in `ARTIFACT_GAPS.md` and must not be described as complete.

4. Figures (separate high-resolution files for journal upload)

Figure	Source
Ground-truth vs. prediction field snapshots (multiple horizons)	Not included: cross-condition arrays/checkpoints and a compatible tutorial T-FNO/U-Net checkpoint pair are unavailable locally
Dry-area-fraction-over-time trace	Phase 4 rollout evaluation output
Pareto-front scatter (interface RMSE vs. mass MAE, all model variants, with bootstrap error bars)	Generated during manuscript polish phase
Lambda-sensitivity curve	Generated during divergence-penalty sensitivity analysis
Loss curves (training/validation, all models/seeds)	Generated at each training phase

Export each at minimum 300 DPI (line-art figures like the sensitivity curve and Pareto front should be vector format — PDF or EPS — if the journal accepts it, since these will need to scale cleanly in print).

5. Tables for Supplementary Material (beyond what fits in the main manuscript)

- Full per-seed results (not just aggregate mean/CI) for every statistical comparison reported in the manuscript — reviewers may specifically request these to spot-check your bootstrap/sign-flip computations.
- Full lambda_div sensitivity table (all five tested values, all three guard metrics).
- Full computational-cost table (already compiled) with per-seed training times, not just means.

6. Statistical audit trail

Explicitly document, in a short supplementary note: the correction made to the sign-flip/Holm implementation (removal of an unnecessary Monte Carlo add-one correction, exclusion of compute-only metrics from the Holm family) and confirmation that all numbers reported in the final manuscript reflect the corrected implementation, not the earlier uncorrected one. This preempts a reviewer question and demonstrates the same transparency already present throughout the project.

7. What NOT to include in the direct journal upload (host externally instead)

- Raw model checkpoints beyond what's needed to verify one representative result per major claim (host the complete set on Zenodo/GitHub, link don't upload, if journal file-size limits are a concern).
- Full per-epoch logs for all 20+ training runs (summarize in the repository, don't inflate the journal's own supplementary PDF).

8. Final pre-submission verification pass

Before uploading anything: 1. Click every link in the manuscript and supplementary material yourself, from a fresh/incognito browser session, to confirm nothing is broken or access-restricted. 2. Re-run at least one full result end-to-end from the public repository (ideally on a clean machine or fresh cloud instance) to confirm the reproducibility instructions actually work as written — this is the single most common reason reviewers flag reproducibility concerns, and it's fully within your control to catch before submission. 3. Confirm every SHA-256 checksum quoted in the manuscript matches the actual file in the archived repository.

Statistical Audit Note

The final stored analyses use one audited implementation for paired inference. For up to 20 seeds, the two-sided sign-flip test enumerates all sign assignments exactly and divides the number at least as extreme as the observed statistic by the exact enumeration size. It does not apply a Monte Carlo add-one correction.

Holm-Bonferroni families exclude compute-only quantities such as parameter count, latency, and throughput. Those quantities remain descriptive. The reviewer-facing script `scripts/reproduce_reported_results.py` independently recomputes paired means and exact sign-flip probabilities from the stored per-seed results and checks the archived bootstrap intervals and Holm-adjusted values against the manuscript.

The cross-condition comparison has five paired seeds. Its minimum attainable two-sided exact sign-flip probability is 0.0625. Accordingly, the cross-condition directions are reported as descriptive and underpowered, not confirmatory.

Recorded Runtime Environment

The final local BubbleML environment was queried during package assembly:

- Python 3.12.7
- PyTorch 2.13.0
- neuraloperator 2.0.0
- NumPy 2.5.1
- h5py 3.16.0
- Matplotlib 3.11.1
- SciPy 1.18.0
- TensorLy 0.9.0
- opt_einsum 3.4.0
- pytest 9.1.1

The tutorial-split benchmark artifacts record Apple M2/MPS. The compact cross-condition summary records CUDA on an NVIDIA Tesla T4. These hardware records are not silently treated as interchangeable.

Reproducibility Self-Test

Initial test date: 2026-08-04. Result: PASS.

Final-audit rerun (2026-08-08)

From the current repository source, the documented command `python scripts/reproduce_reported_results.py --output-dir reproduced` completed with `PASS`. It recomputed the retained tutorial paired statistics and the lambda=.30 non-inferiority check. The full test command `python -m pytest -q` executed 38 tests with 38 passed and no skips or expected failures reported.

This rerun remains a stored-result test, not an assertion of raw-data training reproducibility.

Reviewer simulation

A local commit was cloned into a newly created directory and executed with a new Python 3.12.7 virtual environment. The test used only the public command in the README and no installed third-party packages.

`` `bash git clone /Users/sbmahafujbondhon/Documents/DD-PINNs/bubbleml-submission /private/tmp/bubbleml-reviewer.ssGUhj/repo /Users/sbmahafujbondhon/Documents/DD-PINNs/.venv/bin/python -m venv /private/tmp/bubbleml-reviewer.ssGUhj/venv cd /private/tmp/bubbleml-reviewer.ssGUhj/repo /usr/bin/time -p /private/tmp/bubbleml-reviewer.ssGUhj/venv/bin/python scripts/reproduce_reported_results.py --output-dir reproduced ` ``

Wall time for the reviewer-facing reproduction command was 0.06 seconds (`user 0.04`, `sys 0.00`) on the assembly machine. This time excludes cloning and virtual-environment creation.

Reproduced results

The clean clone independently recomputed from stored per-seed JSON:

- Tutorial T-FNO minus U-Net interface-temperature RMSE difference:

`-0.50189749`, 95% archived bootstrap interval `[-0.74065434, -0.28300856]`, exact two-sided sign-flip p=`0.001953125`, Holm-adjusted p=`0.03515625`.

- Tutorial T-FNO minus U-Net mass-conservation MAE difference:

`+0.04493574`, interval `[+0.03852948, +0.05122263]`, exact p=`0.0009765625`, Holm-adjusted p=`0.02148438`.

- Divergence hybrid (`lambda_div=0.30`) mass-conservation mean `0.09373474`

versus U-Net `0.16585691`; difference `-0.07212217`, archived interval `[-0.07939724, -0.06534623]`, exact one-sided p=`0.000488281`, and the archived non-inferiority decision `True`.

- `reproduced/fig2_dry_area_trace.svg`, SHA-256

`f75383f1e62b288be6561220fd434b45f41d3506b44eadf161b8bab3a5200e10`.

Discrepancies

None. The generated SVG and text report were byte-identical to the committed reference outputs (`git status --short` returned no changes).

Scope

This is a stored-evidence reproducibility test, not a raw-data retraining test. The full ML test suite was separately run in the recorded pinned environment and passed 38/38 tests. A full raw-data-to-checkpoint reviewer test remains blocked by the external data/checkpoint and cross-condition export gaps declared in `ARTIFACT_GAPS.md`; this limitation is not concealed by the PASS above.

Recorded SHA-256 Checksums

These values come from the final manuscript, phase reports, and direct local verification of four representative checkpoints. Files not present in this repository are marked external and cannot be re-verified from this checkout.

Tutorial 48 x 48 source files

File	SHA-256	Status
`Twall-103.hdf5`	`25b305661dee59cf5df49eb2563a4ed08f79664ba377e57877fcda5a948956dd`	external; manuscript train source
`Twall-106.hdf5`	`4308e5be884c3edcf301e9c12bff8d499d53729be3820aa18e255a5152fbd004`	external; manuscript validation source
`Twall-100.hdf5`	`5bf2539628c3595f39517c466f6971da76f137358771592c14ad1a5881e4d1bf`	external; manuscript test source

Legacy native-resolution archive and members

File	SHA-256
`pool-boiling-subcooled-fc72-2d.tar.gz`	`2eba140d74cbb7b01a55f0684227dea94306aea516a0f864831485d31d25f655`
`Twall-79.hdf5`	`c550dab2e2d5db4ecb7ccc159aefba2a43ab61c50d6d22a0a5322d2a45ad07df`
`Twall-81.hdf5`	`16b6cae48b50cb06a202e0f2c56585caf66eb6919e060b61dc7902026ab84452`
`Twall-85.hdf5`	`4f761caf8efcd66f93bcb55c36eea72876d83f85b8b30526a1b48dfeeb7e8dda`
`Twall-90.hdf5`	`35c859305795fda3a59ea16343d75a611ef4e89c7bacfa05c273b47bcc2d7240`
`Twall-95.hdf5`	`75809930d09d013024a76c0695f28f02c6785183e6d402285486fba334b65338`
`Twall-98.hdf5`	`1c3e4f23908ce6137d88abb8beb438ac08996ffbf191c6d91935d606bafdeb15`
`Twall-100.hdf5`	`8529a1b55613449dbbf99c936ce5f8308abfac92fe856891a9b337bd7f2e949e`
`Twall-103.hdf5`	`4ebc2f95c61fa64b4422f51603a35f19a55add15093fdee435db0558d4d458e1`
`Twall-106.hdf5`	`1b035a9cd8c9b765035bfe88ba2993aee96c52e5df6a8e2c4d32f51b8ca8db45`
`Twall-110.hdf5`	`874a07784d0120b402438d69bf66dd04191bd76ac1eb4b8151ad2ddec55134fd`

The same Twall names have different hashes between tutorial and legacy releases because they are different released files/resolutions. They are not interchangeable.

Representative checkpoints

Checkpoint	Bytes	SHA-256	Verification
`phase1_gpu_decisive/tfno_seed_9999.pt`	4,185,773	`f85e4b3811e721abed3ede47b3b9701b2525aad36c60ddb59d766c7fb0738bab`	direct local hash; also recorded in Phase 4

Checkpoint	Bytes	SHA-256	Verification
`phase1_gpu_decisive/unet_seed_7.pt`	31,106,427	`58173aca6b36dbcb2b748a81549435b7288bbbd1cb91dacf97a25d9746feb3d8`	direct local hash; also recorded in Phase 4
`lambda_sensitivity_030/hybrid_div_seed_42.pt`	4,781,797	`b80f5a23aaf13a88f536bee9e95ce85b8fcee1af62a6a4c6916c0ffcd06c7fb6`	direct local hash during submission assembly
`lambda_sensitivity_030/hybrid_div_seed_9999.pt`	4,782,129	`ed5048981a6ff6252cbdec265733de49b9abf019c9f1099bd0f9605851edfcf9`	direct local hash during submission assembly

Other recorded artifact hashes

SHA-256	Artifact	Provenance / verification
`7cfd11a642168b2c845addf79cdfba2191664b1a4f5096ea07149753b7e58b43`	`benchmark_results/multitraj96/report_summary.json`	Direct local SHA-256 recomputation; compact audited 96×96 cross-condition summary.
`492057863f155b49ab2d249dfdacbf6d855aa2fed8882185794cb5ce2cd8d2c6`	`benchmark_results/multitraj384_micro/native384_summary.json`	Direct local SHA-256 recomputation; native-384 one-epoch feasibility summary.
`7c4f8e743b89543c7e8a009f01a42f88b2ece7abd981623cbed02731dd100154`	Unassigned, provenance unrecoverable from this checkout	Phase history identifies it only as a temporary cloud upload bundle at `/private/tmp/ddpinns-future-work.tar.gz`; the bundle bytes are absent, so no local hash confirmation or archival role is claimed.

Artifact Gaps

This file distinguishes available evidence from claims in the supplied manifest that could not be verified in the local project.

Missing from the assembled checkout

- Raw BubbleML HDF5 files and the 10.35 GiB legacy archive. These are intentionally

external, but a final official download URL must be added by the author.

- The complete model checkpoint set. Four representative checkpoint hashes were

verified; checkpoints remain external because the local set is approximately 1.8 GiB.

- Complete per-seed configs, histories, and checkpoint provenance for the 96 x 96

cloud cross-condition experiment. Only the compact audited summary `benchmark_results/multitraj96/report_summary.json` is present.

- Cross-condition field-snapshot arrays/checkpoints needed to regenerate the

ground-truth-versus-prediction snapshot figure. A repository-wide search also found no compatible tutorial-split T-FNO/U-Net checkpoint pair, so the allowed tutorial-split fallback cannot be generated honestly. No substitute is fabricated.

- A source bibliography or citation-key mapping for every prose reference in the

manuscript.

- A verified public GitHub repository/release and archive DOI. The author has

supplied the intended GitHub URL and root MIT `LICENSE`, but the URL was not publicly verified during this audit and no archive DOI has been issued.

Conflicting historical records

An early `phase_history/phase1_report.md` lists a different tutorial split from the final manuscript. This package follows the supplied final manuscript: Twall-103 train, Twall-106 validation, Twall-100 test. Historical files remain unchanged for auditability.

Consequence

The stored-result statistical self-test is complete and independently runnable. A full raw-data-to-checkpoint reproduction is not yet reviewer-runnable without the external files and links listed above.

Bibliographic Verification

Checked on 2026-08-08 against primary publication records. Only works identifiable from the manuscript text are included. A complete citation pass cannot be finished without the authors' intended source list for generic prose allusions.

Verified records

1. Sheikh Md Shakeel Hassan, Arthur Feeney, Akash Dhruv, Jihoon Kim, Youngjoon Suh, Jaiyoung Ryu, Yoonjin Won, and Aparna Chandramowlishwaran. "BubbleML: A Multiphase Multiphysics Dataset and Benchmarks for Machine Learning." Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track, 2023. DOI: 10.52202/075280-0021. Primary record: https://papers.neurips.cc/paper_files/paper/2023/hash/01726ae05d72ddba3ac784a5944fa1ef-Abstract-Datasets_and_Benchmarks.html 2. Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. "Fourier Neural Operator for Parametric Partial Differential Equations." ICLR 2021. Primary record: https://openreview.net/forum?id=c8P9NQVtmnO 3. Olaf Ronneberger, Philipp Fischer, and Thomas Brox. "U-Net: Convolutional Networks for Biomedical Image Segmentation." arXiv:1505.04597, 2015. Primary record: https://arxiv.org/abs/1505.04597 4. Alasdair Tran, Alexander Mathews, Lexing Xie, and Cheng Soon Ong. "Factorized Fourier Neural Operators." ICLR 2023. Primary record: https://openreview.net/forum?id=tmIiMPl4IPa 5. Md Ashiqur Rahman, Zachary E. Ross, and Kamyar Azizzadenesheli. "U-NO: U-shaped Neural Operators." Transactions on Machine Learning Research, 2023. Primary record: https://openreview.net/forum?id=j3oQF9coJd 6. Sheikh Md Shakeel Hassan, Xianwei Zou, Akash Dhruv, and Aparna Chandramowlishwaran. "Bubbleformer: Forecasting Boiling with Transformers." Advances in Neural Information Processing Systems 38, Datasets and Benchmarks Track, 2025. Primary record: https://proceedings.neurips.cc/paper_files/paper/2025/hash/18d2e6c115da385ff6a7ac53655ae95e-Abstract-Datasets_and_Benchmarks_Track.html 7. Novak Zuber. "Hydrodynamic Aspects of Boiling Heat Transfer." AECU-4439, United States Atomic Energy Commission, 1959. DOI: 10.2172/4175511. Primary repository record: https://digital.library.unt.edu/ark:/67531/metadc872491/ 8. Christy Dunlap, Hari Pandey, Stephen Pierson, Daniel Curl, Braden Stevens, Mohammad Ishraq Hossain, Annapurna Parjuli, Chinmaya Joshi, and Han Hu. "Open Multimodal Datasets and Open-Source Software for Data-Driven Modeling of Multiphase Transport and Thermal Systems." arXiv:2605.23037, 2026. Primary record: https://arxiv.org/abs/2605.23037 9. Madhumitha Ravichandran, Guanyu Su, Chi Wang, Jee Hyun Seong, Artyom Kossolapov, Bren Phillips, Md Mahamudur Rahman, and Matteo Bucci. "Decrypting the Boiling Crisis through Data-Driven Exploration of High-Resolution Infrared Thermometry Measurements." Applied Physics Letters 118(25):253903, 2021. DOI: 10.1063/5.0048391. Primary publisher record supplied with the task. 10. Madhumitha Ravichandran, Artyom Kossolapov, Gustavo Matana Aguiar, Bren Phillips, and Matteo Bucci. "Autonomous and Online Detection of Dry Areas on a Boiling Surface Using Deep Learning and Infrared Thermometry." Experimental Thermal and Fluid Science 145:110879, 2023. DOI: 10.1016/j.expthermflusci.2023.110879. Publisher record: https://www.sciencedirect.com/science/article/pii/S0894177723000353 11. Marimuthu Kalimuthu, David Holzmüller, and Mathias Niepert. "LOGLO-FNO: Efficient Learning of Local and Global Features in Fourier Neural Operators." arXiv:2504.04260, 2025. Primary record: https://arxiv.org/abs/2504.04260 12. Jack Richter-Powell, Yaron Lipman, and Ricky T. Q. Chen. "Neural Conservation Laws: A Divergence-Free Perspective." Advances in Neural Information Processing Systems 35, 2022. Primary record: https://arxiv.org/abs/2210.01741 13. Xigui Li, Hongwei Zhang, Ruoxi Jiang, Deshu Chen, Chensen Lin, Limei Han, Yuan Qi, Xin Guo, and Yuan Cheng. "Project and Generate: Divergence-Free Neural Operators for Incompressible Flows." arXiv:2603.24500, 2026. Primary record: https://arxiv.org/abs/2603.24500

Corrected supplied-text discrepancy

The Data Availability Statement calls the 2023 work "A Comprehensive Multi-Physics Dataset..." The verified title is "A Multiphase Multiphysics Dataset and Benchmarks for Machine Learning." The supplied Markdown is retained unchanged; the generated bibliography uses the verified title.

[VERIFY] citations not safely identifiable

- No unresolved generic semi-empirical CHF-correlation allusion remains: the

manuscript was reworded to avoid implying an uncited specific correlation.

- The generic question of which other experimental margin-to-CHF studies the

authors may wish to cite remains outside the two verified infrared/dry-area records above.

These references must remain `[VERIFY]` until the author supplies citation keys or source URLs. No entries were inferred from topic similarity.

Dataset-license check

The current dataset card under the official HPCForge account labels BubbleML as CC BY 4.0: https://huggingface.co/datasets/hpcforge/BubbleML. This verifies the current card, not that the authors' local copy came from the same release. The supplied author statement therefore retains its explicit license-confirmation placeholder until the author confirms the exact downloaded source/version.

Journal-format check

The current official guides for both International Journal of Heat and Mass Transfer and Applied Thermal Engineering permit any consistent reference style at initial submission and require complete bibliographic elements. Applied Thermal Engineering explicitly describes numbered square-bracket citations. Because the final journal has not been chosen, the generated manuscript uses a consistent numbered bibliography and flags missing citations rather than claiming a final journal-specific style.

Figure Quality Audit

Audit date: 2026-08-08.

Each generated line-art figure is supplied as vector PDF and SVG plus a 600 dpi PNG. Vector files are the preferred submission versions. All six families were rasterized at reduced size and inspected for readable labels, non-overlapping legends, and visible markers.

Figure family	Evidence source	Output dimensions/status
`fig1_pareto_front`	Stored tutorial and lambda-sensitivity result JSON	PDF/SVG; PNG 6000 x 4000; ready
`fig2_dry_area_trace`	Stored Phase 4 rollout series	PDF/SVG; PNG 6000 x 4000; ready
`fig3_lambda_sensitivity`	Stored lambda-sensitivity results	PDF/SVG; PNG 9000 x 3000; ready
`fig4_loss_curves`	Available stored experiment histories	PDF/SVG; PNG 7500 x 5500; ready
`fig5_benchmark_workflow`	Frozen protocol and declared artifact boundary	PDF/SVG; PNG 9000 x 3000; ready
`fig6_split_design`	Frozen tutorial and cross-condition split definitions	PDF/SVG; PNG 8000 x 3750; ready

The requested ground-truth-versus-prediction field-snapshot figure is not in this package because the necessary cross-condition arrays/checkpoints were not available. The permitted tutorial-split fallback also cannot be generated because no compatible local T-FNO/U-Net checkpoint pair survives. It is explicitly marked as missing rather than synthesized.